

Digital Electronics (II SEM B.Tech. CSE)

Couse Code: NECE102



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Digital Electronics: Course Goals and Key Takeaways



Course Objective

To develop an understanding of digital electronics and its application which will be further useful to in the advanced courses of Electronics and Industrial Engineering.

Learning Outcomes

Upon successful completion of this course, students will:

- Gain information about the basic knowledge of digital electronics to design various types of digital circuits and systems.
- Learn how to analyse the performance and working of digital circuits.
- Develop their knowledge of digital electronics for industrial and robotic applications.

Digital Electronics: Key Topics at a Glance



Unit No.	Topics to be Covered	Lecture Hours	Learning Outcome
1.	Number System, Logic Gates, Boolean Algebra, NAND and NOR Implementation of Logic Functions, Determination of Boolean Functions from Digital Circuits.	08	Acquire an understanding of number system, basic logic gates and their applications to develop digital circuits and systems.
2.	Standard Representations of Logical Functions, SOP and POS form, Simplification of Logical Functions using K-map.	08	Acquire the knowledge to represent the logic functions in standard form and its minimization.
3.	Combinational Logic Circuit: Adder, Subtractor, Magnitude Comparator, Code Converters etc. MSI Circuits: Multiplexers, Demultiplexers, Decoders, Encoders etc.	08	Understand the design and applications of combinational logic circuits.
4.	Sequential Circuits: Latches and Flip-flops, Counters, Shift Registers.	10	Acquire the ability to design and perform the analysis of different sequential circuits.
5.	Analysis of Sequential circuits: State tables and state diagrams, Timing Circuits	04	Acquire an understanding on the analysis of sequential circuits.
6.	Introduction to logic families, semiconductor memories, ROM, PAL, PLA and Microprocessor	04	Develop knowledge about the various techniques for IC designing for digital circuits.

Textbook:	Reference Books:
1. Floyd T.L., Digital Fundamentals, 10/2, Pearson Education, 2011.	1. Digital Logic and Computer Design by M. Morris Mano, Prentice –Hall, 2015. 2. Fundamental of Digital Electronics by A. Anand Kumar, 4/e Prentice Hall, 2016.

Conversion of Number System



Problem 1 : Convert $(1011011101101110)_2$ to Decimal

Answer: $(46958)_{10}$

$$\begin{aligned} 1011, 0111, 0110, 1110_2 &= B76E_{16} \\ B76E_{16} &= (11 \times 16^3) + (7 \times 16^2) + (6 \times 16^1) + (14 \times 16^0) \\ &= 45056 + 1792 + 96 + 14 \\ &= 46958_{10} \end{aligned}$$

Octal to Hexadecimal Conversion

To convert an octal number to hexadecimal number, the simplest way is to first convert the given octal number to binary and then binary to hexadecimal.

Problem 1 : Convert $(756.603)_8$ to Hexadecimal

Answer: $(1EE.C18)_{16}$

Given octal number is	7	5	6	.	6	0	3
Convert each octal digit to binary	111	101	110	.	110	000	011
Groups of four bits are	0001	1110	1110	.	1100	0001	1000
Convert each four-bit group to hex	1	E	E	.	C	1	8
The result is	$1EE.C18_{16}$						

Conversion of Number System



Hexadecimal to Octal Conversion

To convert a hexadecimal to octal number, the simplest way is to first convert the given hexadecimal number to binary and then binary to octal.

Problem 1 : Convert $(B9F.AE)_{16}$ to Octal

Answer: $(5637.534)_8$

Given hex number is

Convert each hex digit to binary

Groups of three bits are

Convert each three-bit group to octal

The result is

	B	9	F	.	A	E	
	1011	1001	1111	.	1010	1110	
	101	110	011	111	.	101	011 100
	5	6	3	7	.	5	3 4
	5637.534						

Number System



Binary Addition

Rules for the binary addition:

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 10 \text{ i.e., } 0 \text{ with a Carry of } 1$$

Problem 1 : Add the binary numbers 1101.101 and 111.011.

Answer: 10101.000

8	4	2	1		2^{-1}	2^{-2}	2^{-3}	(Column numbers)
1	1	0	1	.	1	0	1	
+	1	1	1	.	0	1	1	
1	0	1	0	1	.	0	0	0

In the 2^{-3} 's column

$1 + 1 = 0$, with a carry of 1 to the 2^{-2} column

In the 2^{-2} 's column

$0 + 1 + 1 = 0$, with a carry of 1 to the 2^{-1} column

In the 2^{-1} 's column

$1 + 0 + 1 = 0$, with a carry of 1 to the 1's column

In the 1's column

$1 + 1 + 1 = 1$, with a carry of 1 to the 2's column

In the 2's column

$0 + 1 + 1 = 0$, with a carry of 1 to the 4's column

In the 4's column

$1 + 1 + 1 = 1$, with a carry of 1 to the 8's column

In the 8's column

$1 + 1 = 0$, with a carry of 1 to the 16's column

Number System



Binary Subtraction

Rules for the binary subtraction:

$$0 - 0 = 0$$

$$1 - 1 = 0$$

$$1 - 0 = 1$$

$$0 - 1 = 1 \text{ with a borrow of 1}$$

Problem 1 : Subtract $(111.11)_2$ from $(1010.01)_2$.

Answer: 10101.000

8	4	2	1		2^{-1}	2^{-2}	2^{-3}	
1	0	1	0	.	0	1	0	
		1	1	.	1	1	1	
0	0	1	0	.	0	1	1	

(Column numbers)

In the 2^{-3} column, a 1 cannot be subtracted from a 0. So, borrow a 1 from the 2^{-2} column making the 2^{-2} column 0. The 1 borrowed from the 2^{-2} column becomes 10 in the 2^{-3} column. Therefore, in the 2^{-3} column, $10 - 1 = 1$.

In the 2^{-2} column, a 1 cannot be subtracted from a 0. So, borrow a 1 from the 2^{-1} column, but it is also a 0. So, borrow a 1 from the 1's column. That is also a 0, so borrow a 1 from the 2's column making the 2's column 0. This 1 borrowed from the 2's column becomes 10 in the 1's column. Keep one 1 in the 1's column, bring the other 1 to the 2^{-1} column, which becomes 10 in this column. Keep one 1 in the 2^{-1} column and bring the other 1 to the 2^{-2} column, which becomes 10 in this column. Therefore,

$$\text{In the } 2^{-2}\text{'s column} \quad 10 - 1 = 1$$

$$\text{In the } 2^{-1}\text{'s column} \quad 1 - 1 = 0$$

$$\text{In the 1's column} \quad 1 - 1 = 0$$

Number System



Binary Multiplication

Rules for the binary subtraction:

$$0 \times 0 = 0$$

$$1 \times 1 = 1$$

$$1 \times 0 = 0$$

$$0 \times 1 = 0$$

Problem 1 : Multiply $(1101)_2$ from $(110)_2$.

Answer: 1001110

$$\begin{array}{r} 1101 \\ \times 110 \\ \hline 0000 \\ 1101 \\ 1101 \\ \hline 1001110 \end{array}$$

Problem 1 : Multiply $(1011.101)_2$ from $(101.01)_2$.

Answer: 111101.00001

$$\begin{array}{r} 1011.101 \\ \times 101.01 \\ \hline 1011101 \\ 0000000 \\ 1011101 \\ 0000000 \\ 1011101 \\ \hline 111101.00001 \end{array}$$

Number System



Binary Division

Problem 1 : Divide $(101101)_2$ by $(110)_2$.
Answer: 111.1

$$\begin{array}{r} 110 \overline{) 101101 (111.1} \\ \underline{110} \\ 1010 \\ \underline{110} \\ 1001 \\ \underline{110} \\ 110 \\ \underline{110} \\ 000 \end{array}$$

Problem 1 : Divide $(110101.11)_2$ by $(101)_2$.
Answer: 1010.11

$$\begin{array}{r} 101 \overline{) 110101.11 (1010.11} \\ \underline{101} \\ 0110 \\ \underline{101} \\ 111 \\ \underline{101} \\ 101 \\ \underline{101} \\ 000 \end{array}$$

Number System



r's Complement

- Complements are used in digital computers to simplify the subtraction operation.

1. To perform subtraction using addition

Subtraction is harder to implement in digital circuits than addition.

Using **complements**, subtraction can be converted into addition:

$$A - B = A + (\text{complement of } B)$$

Example (decimal – 10's complement):

$$456 - 178 = 456 + (10^3 - 178) = 456 + 822 = 1278$$

Discard the carry → **278**

This is much easier for computers than direct subtraction.

N = Given Number

n = Total Number of Digits in the given number

r = Base

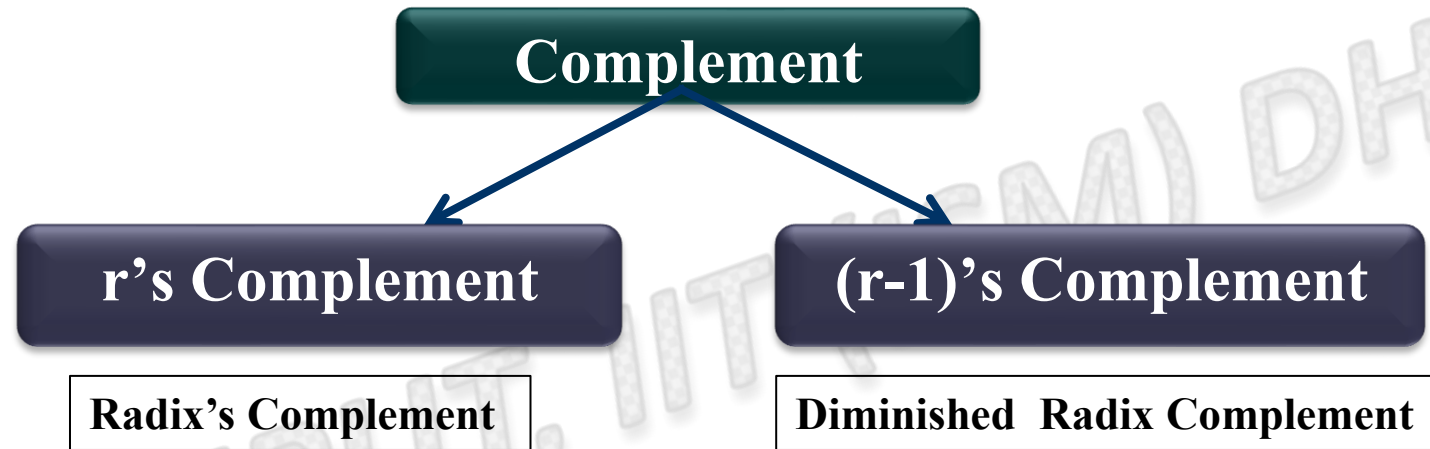
$r^n - N \rightarrow$ Formula for r's Complement

Number System



r's Complement

- Complements are used in digital computers to simplify the subtraction operation.



Example: Complement of $(7)_{10}$
Answer: $10-7=3$

N = Given Number

n = Total Number of Digits in the given number

r = Base

$r^n - N \rightarrow$ Formula for r's Complement

Number System



Example: $N = 5690$; Determine 10's Complement

Answer: $N = 5690$, $n = 4$, $r = 10$

$$10^4 - 5690 = 10000 - 5690 = 4310$$

Example: $N = 1101$; Determine 2's Complement

Answer: $N = 1101$, $n = 4$, $r = 2$

$$2^4 - 1101 = (16)_{10} - 1101 = 10000 - 1101 = 11$$

Example: $N = 76895$; Determine 10's Complement

Answer: 23015

Example: $N = 11011$; Determine 2's Complement

Answer: 00101

Number System



(r-1)'s Complement

Decimal Number System

$r = 10$;

r 's complement $\rightarrow$ 10's Complement

($r-1$)'s complement $\rightarrow$ 9's Complement

Binary Number System

$r = 2$;

r 's complement $\rightarrow$ 2's Complement

($r-1$)'s complement $\rightarrow$ 1's Complement

Octal Number System

$r = 8$;

r 's complement $\rightarrow$ 8's Complement

($r-1$)'s complement $\rightarrow$ 7's Complement

Hexadecimal Number System

$r = 16$;

r 's complement $\rightarrow$ 16's Complement

($r-1$)'s complement $\rightarrow$ F's (15's) Complement

Calculating r 's complement using ($r-1$)'s Complement

$$r\text{'s complement} = r^n - N$$

$$(r-1)\text{'s complement} = r^n - N - 1$$

$$(r-1)\text{'s complement} = r\text{'s complement} - 1$$

$$r\text{'s complement} = (r-1)\text{'s complement} + 1$$



Borrow operation is involved No Borrow Operation is involved

Example: 7's complement of Octal number 5674;

Answer: $N = 5674$; $r = 8$; $n = 4$

$$(r-1)\text{'s complement} = r^n - N - 1$$

$$(8-1)\text{'s complement} = 8^4 - 5674 - 1 = (4096)_{10} - 5674 - 1 = 10000 - 5674 - 1$$

Number System



(r-1)'s Complement

Example: 7's complement of Octal number 5674;

Answer: $N = 5674$; $r = 8$; $n = 4$

(r-1)'s complement = $r^n - N - 1$

(8-1)'s complement = $7777 - 5674 = 2103$

8's complement = 7's complement + 1 = 2104

Example: 1's complement of 1101.

Answer: $N = 1101$; $r = 2$; $n = 4$

(r-1)'s complement = $r^n - N - 1$

(2-1)'s complement = $1111 - 1101 = 0010$

2's complement = 1's complement + 1 = $0010 + 1 = 0011$

$$(r-1)'s \text{ complement} = r^n - N - 1 = (r^n - 1) - N$$

Example: Obtain 1's complement of 1010.

Answer: $N = 1010$; $r = 2$; $n = 4$

(r-1)'s complement = $r^n - 1 - N$

(2-1)'s complement = $1111 - 1010 = 0101$

2's complement = 1's complement + 1 = $0101 + 1 = 0110$

Number System



(r-1)'s Complement

Example: Obtain 2's complement of 10111010.

Answer: $N = 10111010; r = 2; n = 8$

(r-1)'s complement = $r^n - 1 - N$

(2-1)'s complement = $11111111 - 10111010 = 01000101$

2's complement = 1's complement + 1 = $0101 + 1 = 01000110$

Problem: Obtain 1's complement of 100111

Answer: 011000

Problem: Obtain 2's complement 1100111

Answer: 0011001

Number System



Representation of Signed Numbers

- In signed magnitude form, an additional bit called the signed bit is placed in front of the number.
- If the sign bit is 0, the number is positive, If it is 1, the number is negative.

0	1	0	1	0	0	1
---	---	---	---	---	---	---

Sign bit Magnitude

= + 41

1	1	0	1	0	0	1
---	---	---	---	---	---	---

Sign bit Magnitude

= - 41

Logic Gates



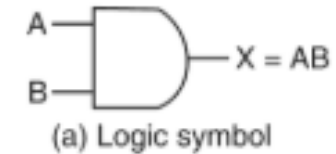
- Logic Gates are the fundamental building blocks of digital systems.
- Name Logic Gate is derived from the ability of such a device to make decisions.
- Produces one output level when some combinations of input levels are present and a different output level when other combinations of input levels are present.
- The fact that computers are able to perform very complex logic operations, stems from the way these elementary gates are interconnected.
- The interconnection of gates to perform a variety of logical operations is called Logic Design.

Basic GATES

AND

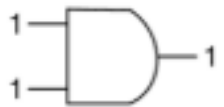
OR

NOT



AND GATE

- AND GATE → 2 or more Inputs but only 1 Output.
- Output assumes the Logic 1 state, only when each of its inputs is at Logic 1 state.
- Output assumes the Logic 0 state, even if one of its inputs is at Logic 0 state.
- AND GATE → Device whose output is 1, if and only if all its inputs are 1. → AND gate → All or Nothing Gate



2- Input AND Gate

Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

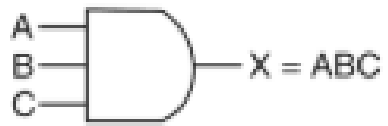
(b) Truth table

Logic Gates



AND GATE

3- Input AND Gate



(a) Logic symbol

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(b) Truth table

OR GATE

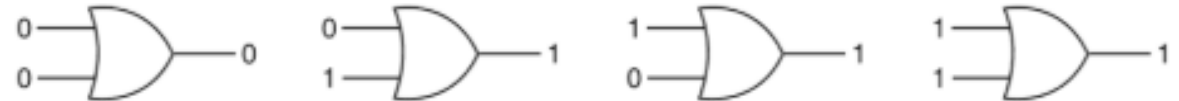
2- Input OR Gate



(a) Logic symbol

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

(b) Truth table



3- Input OR Gate

- **OR GATE → 2 or more Inputs but only 1 Output.**
- **Output assumes the Logic 1 state, even if one of its inputs is in Logic 1 state.**
- **Output assumes the Logic 0 state, only when each one of its inputs is in Logic 0 state.**
- **OR gate → device whose output is 1, even if one of its inputs is 1.**
- **OR Gate is also called an Any or All Gate.**
- **Can also be called Inclusive OR Gate**



(a) Logic symbol

Inputs			Output
A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

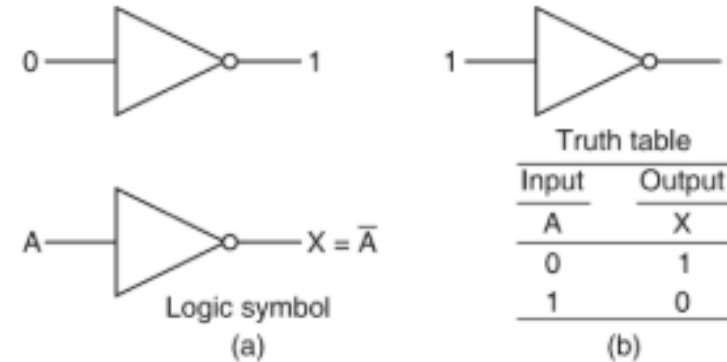
(b) Truth table

Logic Gates



NOT GATE (INVERTER)

- Only one input and of course only one output.
- Device whose output is always the complement of its input.
- Output of a NOT gate assumes the logic 1 state when its input is in logic 0 state and assumes the logic 0 state when its input is in logic 1 state.



Universal GATES

NAND

NOR

- Though logic circuits of any complexity can be realized by using only three basic gates (AND, OR and NOT).
- The 2 universal GATES (NAND and NOR), each of which can also realize logic circuits single-handedly.
- NAND and NOR gates can perform all the three basic logic functions (AND, OR and NOT).

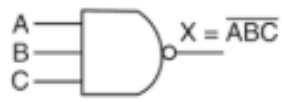
Logic Gates



NAND GATE

- **NAND** means **NOT AND** i.e., the **AND** output is **NOT**ed.
- **NAND** gate is a combination of an **AND** gate and a **NOT** gate.

3- Input NAND Gate

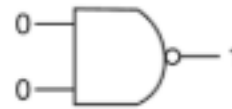


(a) Logic symbol

Inputs			Output
A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

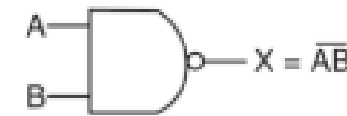
(b) Truth table

2- Input NAND Gate



An AND gate followed by a NOT gate

(a)



A two-input NAND gate (Logic symbol)

Truth table

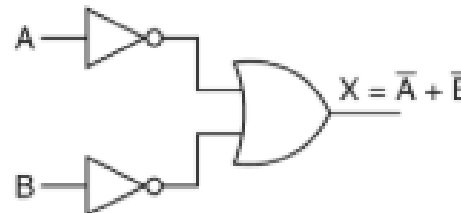
Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

(b)

Bubbled OR Gate:

Bubbled OR Gate:

- **OR** Gate with inverted inputs → **Bubbled OR** Gate.
- **NAND** gate is equivalent to **Bubbled OR** Gate.



(a)



Inputs		Inverted inputs		Output
A	B	A-bar	B-bar	X
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

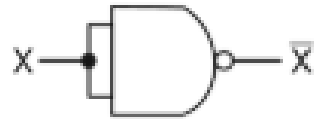
(b) Truth table

Logic Gates

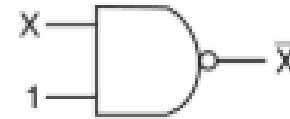


NAND Gate as an Inverter

- NAND gate can also be used as an inverter by tying all its input terminals together and applying the signal to be inverted to the common terminal.
- Or by connecting all its input terminals except one, to Logic 1 and applying the signal to be inverted to the remaining terminal.



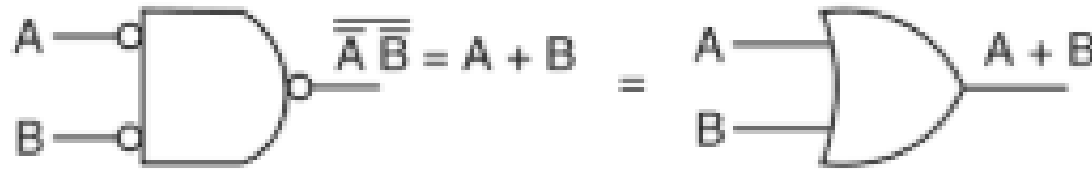
(a) NAND gate as an inverter



(b) NAND gate as a controlled inverter

Bubbled NAND Gate

- Bubbled NAND gate is equivalent to OR gate.



Logic Gates



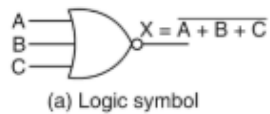
NOR GATE

- NOR means NOT OR i.e., the OR output is NOTed.
- NOR gate is a combination of an OR gate and a NOT gate.

2- Input NOR Gate

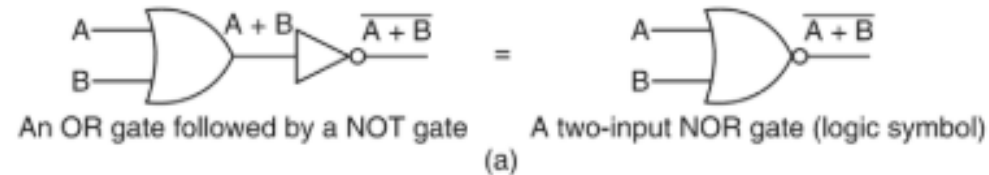


3- Input NOR Gate



Inputs			Output
A	B	C	X
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

(b) Truth table



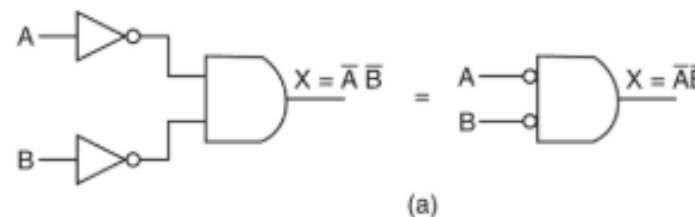
Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

(b) Truth table

Bubbled AND Gate:

- AND Gate with inverted inputs → Bubbled AND Gate.
- NOR gate is equivalent to Bubbled AND Gate.

Bubbled AND Gate:



Inputs		Inverted inputs		Output
A	B	A	B	X
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

(b) Truth table

Logic Gates

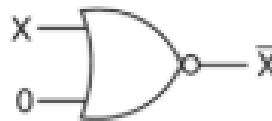


NOR Gate as an Inverter

- NOR gate can also be used as an inverter by tying all its input terminals together and applying the signal to be inverted to the common terminal.
- Or by connecting all its input terminals except one, to Logic 0 and applying the signal to be inverted to the remaining terminal.



(a) NOR gate as an inverter



(b) NOR gate as a controlled inverter

Bubbled NOR Gate

- Bubbled NOR Gate is equivalent to AND gate.

